

Measurement and Visualization of Cryptocurrency Adoption and Market Behavior Using On-Chain Data

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Abstract—Cryptocurrency networks provide transparent and continuously updated records of user activity, transaction behavior, and market performance. This paper presents an analytical framework for measuring cryptocurrency adoption, network activity, and financial behavior using on-chain and market data from Bitcoin and Ethereum. The dataset contains 8,952 daily observations and 15 features collected from Blockchain.com, CoinMetrics, and CoinGecko. Six research questions are addressed using Linear Regression, Random Forest, Gradient Boosting, Logistic Regression, LightGBM, XGBoost, and Hidden Markov Models. Results show that Linear Regression is not effective for return modeling, producing a negative R-squared score. On-chain metrics are also weak predictors of next-day return direction, with the best return classification model achieving only 57.16 percent F1-score. However, the same indicators strongly predict volatility, where Logistic Regression achieves 96.81 percent accuracy and 100 percent recall. LightGBM with SMOTE achieves 93.48 percent accuracy for distinguishing genuine usage from speculative behavior. Transaction fees and hash rate also show strong predictive value for adoption phases, with Random Forest achieving 95.20 percent accuracy for Daily Active Address classification. In contrast, HMM-based market cycle detection and NVT-based valuation show limited reliability.

Index Terms—Cryptocurrency, Blockchain, On-Chain Analytics, Machine Learning, Bitcoin, Ethereum, Network Activity

I. INTRODUCTION

Blockchain networks generate open and verifiable records of transactions, user activity, and network security. Unlike traditional financial systems, cryptocurrency ecosystems provide publicly accessible data that can be analyzed to understand adoption, usage, speculation, and financial performance. From a computer science perspective, this problem requires scalable data processing, time-series analysis, graph-based reasoning, and machine learning methods for extracting useful signals from noisy blockchain data [1].

Despite increasing adoption, several challenges remain. On-chain data are heterogeneous and noisy because address reuse, exchange wallets, protocol-generated change outputs, and wash trading may distort genuine economic activity [2], [3]. Cryptocurrency markets are also highly volatile and non-stationary, making reliable prediction difficult. Prior studies often focus on Bitcoin alone or analyze market and network indicators separately [4], [6]. Therefore, a unified framework is

needed to evaluate how adoption, network activity, and market behavior interact.

This study addresses six research questions: whether active addresses lead to greater transaction activity; which on-chain metrics predict returns and volatility; how transaction count and volume distinguish genuine usage from speculation; how fees and hash rate influence adoption and value; whether liquidity and market capitalization detect market cycles; and whether the NVT ratio identifies overvalued or undervalued assets.

The main contributions are:

- A unified analytical framework for Bitcoin and Ethereum using 8,952 daily observations and 15 on-chain and market features.
- A comparative evaluation of regression, ensemble learning, gradient boosting, resampling-based classification, and HMM methods.
- An empirical demonstration that on-chain metrics strongly support volatility, adoption, and behavior classification, but are limited for short-term return prediction, NVT valuation, and HMM regime detection.

II. RELATED WORK

This review is guided by a prior PRISMA-style systematic review that screened 480 records and retained 50 empirical studies published between 2021 and 2026. The retained studies focused on on-chain metrics, wallet growth, active addresses, transaction activity, transaction fees, consensus participation, volatility, liquidity, and valuation indicators. The review found that market performance and network adoption are the two most common research themes, while data volatility, scalability, privacy, and regulation remain major technical challenges.

Recent studies show that cryptocurrency forecasting improves when market indicators are combined with on-chain variables. Berger and Koubova [5] compared econometric time-series models with machine learning approaches for Bitcoin return forecasting. Guo et al. [6] demonstrated cross-cryptocurrency return predictability using adaptive LASSO. Huang et al. [7] showed that machine learning techniques improve Bitcoin volatility forecasting, while King et al. [4]

analyzed blockchain metrics and indicators for cryptocurrency trading. Zhou et al. [8] further proposed graph neural network-based volatility forecasting, showing the increasing role of advanced machine learning in cryptocurrency analytics.

Other studies emphasize that raw on-chain transaction data must be interpreted carefully. Liu et al. [2] surveyed cryptocurrency transaction knowledge discovery and highlighted the need for standardized evaluation. Cong et al. [3] showed that wash trading can distort observed transaction behavior. Rzayev et al. [10] studied adoption dynamics and transaction activity, while Kim et al. [9] examined whether higher hash rate strengthens Bitcoin network security. Moncada et al. [11] studied instability using the Network Value to Transactions ratio, supporting the need to test NVT as a valuation indicator rather than assuming it is reliable.

The literature shows three main gaps. First, many studies focus only on Bitcoin or Ethereum, limiting generalizability. Second, on-chain and market indicators are often studied separately. Third, valuation measures such as NVT and regime detection methods are rarely compared across multiple model families. This work addresses these gaps using a unified BTC and ETH dataset and six connected research questions.

III. RESEARCH METHODOLOGY

A. Dataset and Preprocessing

The dataset was collected from Blockchain.com, Coin-Metrics, and CoinGecko. It contains 8,952 daily observations for Bitcoin and Ethereum with 15 features: date, ticker, price, volume, market capitalization, transaction count, transaction volume, Daily Active Addresses, NVT, hash rate, transaction fees, volatility, liquidity, and ticker encoding.

No missing values were found. Zero-volume entries in early Bitcoin records were handled using forward fill to preserve temporal continuity. Duplicate records were checked using the date-ticker composite key. The date field was converted to datetime format, tickers were standardized, and feature names were normalized. The ticker column was label encoded as BTC equals 0 and ETH equals 1. Z-score normalization was applied after train-test splitting to prevent data leakage.

TABLE I: Dataset Distribution

Split	Samples	Proportion
Training Set	6,265	70%
Test Set	1,790	20%
Validation Set	896	10%

B. Models Used

For RQ1, Linear Regression was used to examine whether selected on-chain and market indicators can model cryptocurrency return:

$$y = \beta_0 + \beta_1 x + \epsilon \quad (1)$$

where x represents the selected input feature set, y represents the target return, β_0 is the intercept, β_1 is the coefficient, and ϵ is the error term. Linear Regression was selected because

it provides an interpretable baseline for measuring whether a direct linear relationship exists between blockchain activity and return behavior.

For RQ2, Logistic Regression, Random Forest, and Gradient Boosting were used to classify next-day return direction and volatility level from transaction count, transaction volume, fees, hash rate, NVT, and liquidity. Logistic Regression provides a linear baseline, while Random Forest and Gradient Boosting capture non-linear relationships.

For RQ3, LightGBM and XGBoost were combined with SMOTE and SMOTEENN to classify genuine usage and speculative behavior. Genuine usage was labeled as 0, while speculative behavior was labeled as 1. Speculative records represented only 6.23 percent of the dataset, requiring resampling.

For RQ4, Random Forest was used to evaluate the influence of transaction fees and hash rate on adoption and market value. Binary targets were created using a 30-day rolling average:

$$y_t = \begin{cases} 1, & V_t > \frac{1}{n} \sum_{i=1}^n V_{t-i} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where $n = 30$.

For RQ5, a Hidden Markov Model detected hidden bull and bear regimes using market capitalization, liquidity, and volatility. For RQ6, Random Forest, XGBoost, and Logistic Regression were used for NVT-based overvalued and undervalued classification.

C. Performance Metrics

Regression performance was evaluated using coefficient of determination (R^2), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE):

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (3)$$

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2, \quad RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \quad (4)$$

Classification performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN} \quad (6)$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (7)$$

IV. EXPERIMENTAL EVALUATION

A. RQ1: Regression-Based Return Prediction

The performance of the regression model was evaluated using R^2 , MSE, and RMSE on the test dataset. These metrics were computed to assess how well the model predicts the target return based on the selected feature set. The experimental results are shown in Table II.

The obtained R^2 value is negative, indicating that the regression model performs significantly worse than a naive

TABLE II: Linear Regression Results for RQ1

Metric	Value
R^2 Score	-3.2908
MSE	1.113×10^{12}
RMSE	1.055×10^6

baseline model that predicts the mean of the target variable. This suggests that the selected feature representation and linear model configuration are not sufficient to capture the underlying relationship between on-chain variables and cryptocurrency return.

The high RMSE value, approximately 1.055×10^6 , reflects substantial deviation between predicted and actual values. This confirms large prediction errors on the test set and shows the limited predictive capability of the model in its current form. Overall, the RQ1 results demonstrate that Linear Regression is not effective for modeling return behavior in this dataset. The poor performance may be caused by non-linear relationships, outliers, and insufficient feature representation.

B. RQ2: Returns and Volatility Prediction

Table III shows that on-chain metrics are weak predictors of next-day return direction. Random Forest achieved the highest F1-score for returns, but the value remained only 57.16 percent. In contrast, volatility prediction was highly accurate. Logistic Regression achieved 96.81 percent accuracy and 100 percent recall, showing that on-chain metrics are more useful for risk monitoring than price direction prediction.

TABLE III: Model Performance for RQ2

Target	Model	Acc.	Prec.	Recall	F1
Returns	GB	48.71	50.18	45.67	47.81
	RF	49.27	50.54	65.76	57.16
	LR	49.22	50.67	49.24	49.94
Volatility	GB	83.22	75.40	95.22	84.16
	RF	94.74	93.35	95.58	94.45
	LR	96.81	93.62	100.00	96.71

TABLE IV: Confusion Matrices for Best RQ2 Models

Random Forest for Returns		
	Pred. Up	Pred. Down
Actual Up	329	565
Actual Down	652	242
Logistic Regression for Volatility		
	Pred. High	Pred. Low
Actual High	894	0
Actual Low	57	837

C. RQ3: Genuine Usage and Speculative Behavior

RQ3 examined whether transaction count and transaction volume patterns distinguish genuine network usage from speculative behavior. Genuine usage represents normal economic activity, while speculative behavior represents abnormal transaction behavior associated with strong price movement,

unusually large transaction values, and weak transaction count growth. Since speculative records represented only 6.23 percent of the dataset, SMOTE and SMOTEENN were applied to reduce class imbalance.

Table V shows that LightGBM with SMOTE achieved the best practical balance, reaching 93.48 percent accuracy, 60.99 percent F1-score, and 0.890 ROC-AUC. Although LightGBM with SMOTEENN achieved the highest ROC-AUC of 0.907, its lower precision reduced its overall reliability. Therefore, LightGBM with SMOTE was selected as the best final model.

TABLE V: Model Performance for RQ3

Model	Acc.	Prec.	Rec.	F1	ROC	PR
LGBM+SMOTE	93.48	61.26	60.71	60.99	0.890	0.587
LGBM+SMOTEENN	88.08	38.65	71.43	50.16	0.907	0.606
XGB+SMOTEENN	83.51	29.85	71.43	42.11	0.866	0.320
XGB+SMOTE	85.31	30.00	56.25	39.13	0.808	0.235

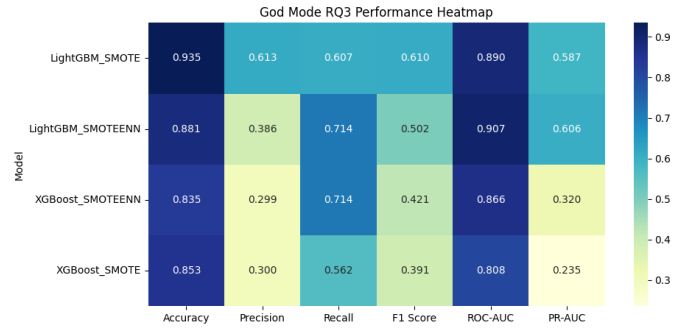


Fig. 1: RQ3 performance heatmap for LightGBM and XGBoost models.

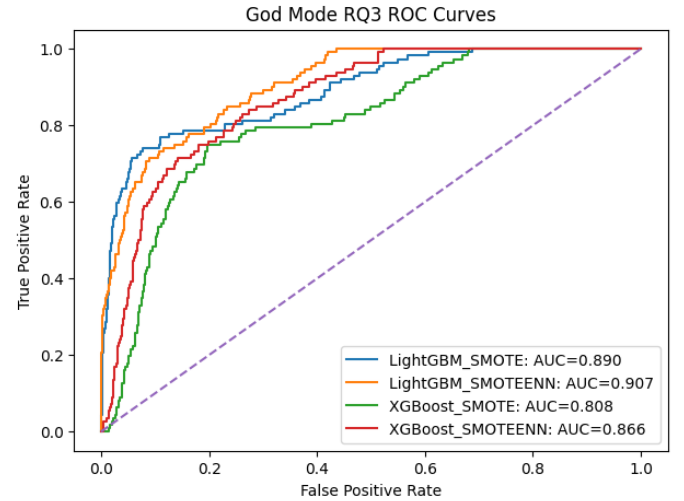


Fig. 2: ROC curves for RQ3 speculative behavior classification.

The SHAP summary in Fig. 3 shows that absolute return, transaction count change, volatility, transaction count, and lagged transaction count were the strongest contributors.

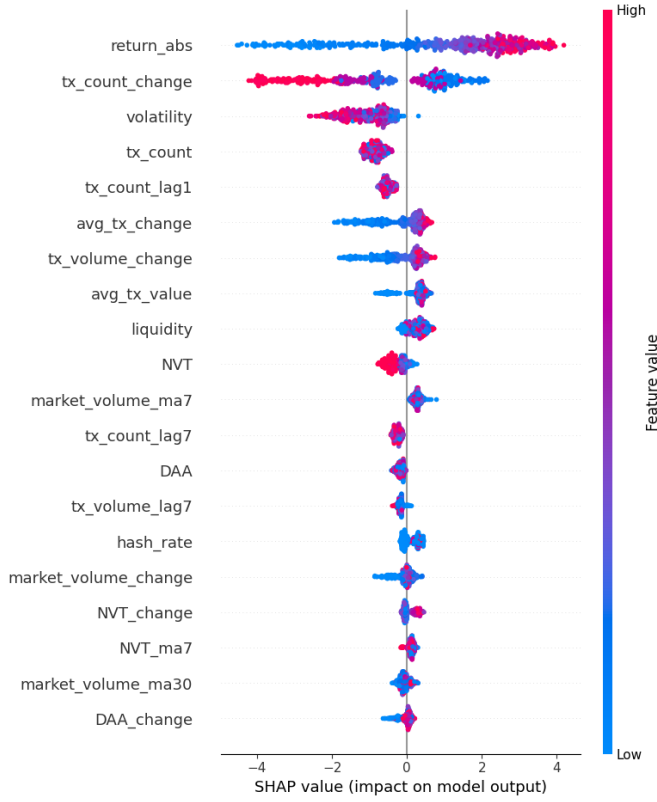


Fig. 3: SHAP feature importance for speculative behavior classification.

This supports the conclusion that speculative behavior is not explained by a single variable, but by the interaction of transaction intensity, price movement, and volatility.

D. RQ4: Fees, Hash Rate, Adoption, and Value

For RQ4, Random Forest was used to evaluate how transaction fees and consensus participation influence adoption and market value. Table VI shows strong classification performance for both targets. DAA adoption achieved 95.20 percent accuracy and 95.36 percent F1-score, while market capitalization achieved 93.63 percent accuracy and 93.69 percent F1-score.

TABLE VI: Random Forest Performance for RQ4

Target	Acc.	Prec.	Recall	F1
DAA	95.20	95.87	94.84	95.36
Market Cap	93.63	93.90	93.48	93.69

Fig. 4 shows that DAA and transaction count have a very strong positive correlation of 0.90, confirming that active address growth is closely connected to real network usage. Market capitalization also shows moderate positive relationships with fees, hash rate, and DAA, suggesting that adoption and network security indicators contribute to value formation.

TABLE VII: Confusion Matrices for RQ4

DAA Adoption		
	Pred. 0	Pred. 1
Actual 0	821	38
Actual 1	48	883

Market Capitalization		
	Pred. 0	Pred. 1
Actual 0	830	55
Actual 1	59	846

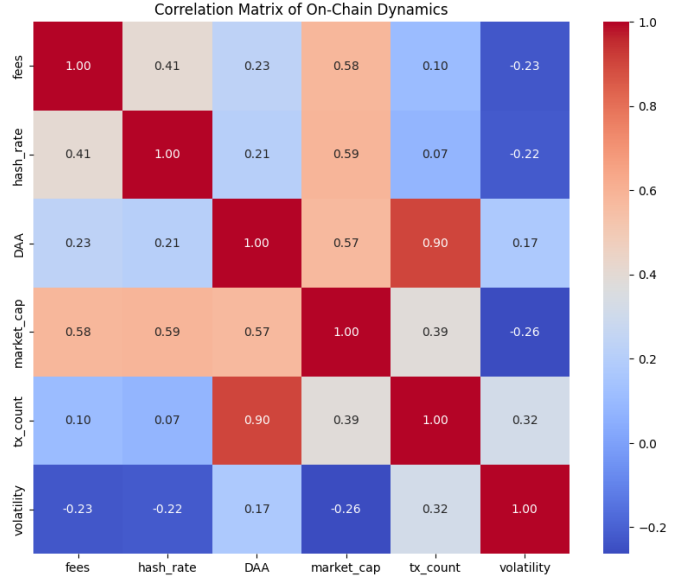


Fig. 4: Correlation matrix of fees, hash rate, DAA, market capitalization, transaction count, and volatility.

E. RQ5 and RQ6: Market Cycles and NVT Valuation

The HMM results show weak generalization for market regime detection. Although the training set produced moderate performance, the test set classified all observations as bear regime, resulting in zero bull-regime recall and an F1-score of 0.000. This suggests that market capitalization, liquidity, and volatility alone are insufficient for stable market cycle detection.

TABLE VIII: RQ5 and RQ6 Results

RQ	Model/Split	Acc.	Prec.	Recall	F1
RQ5	HMM Train	0.523	0.662	0.387	0.489
	HMM Test	0.458	0.000	0.000	0.000
	HMM Val.	0.494	1.000	0.002	0.005
RQ6	RF	0.482	0.222	0.312	0.259
	XGBoost	0.486	0.224	0.510	0.311
	LR	0.546	0.235	1.000	0.380

Fig. 5 explains the poor HMM performance. The model detected both bear and bull regimes during training, but collapsed toward bear-regime predictions on the test and validation sets. This indicates that the selected market variables provide unstable regime boundaries. For RQ6, Logistic Regression performed best among the NVT-based valuation models, but

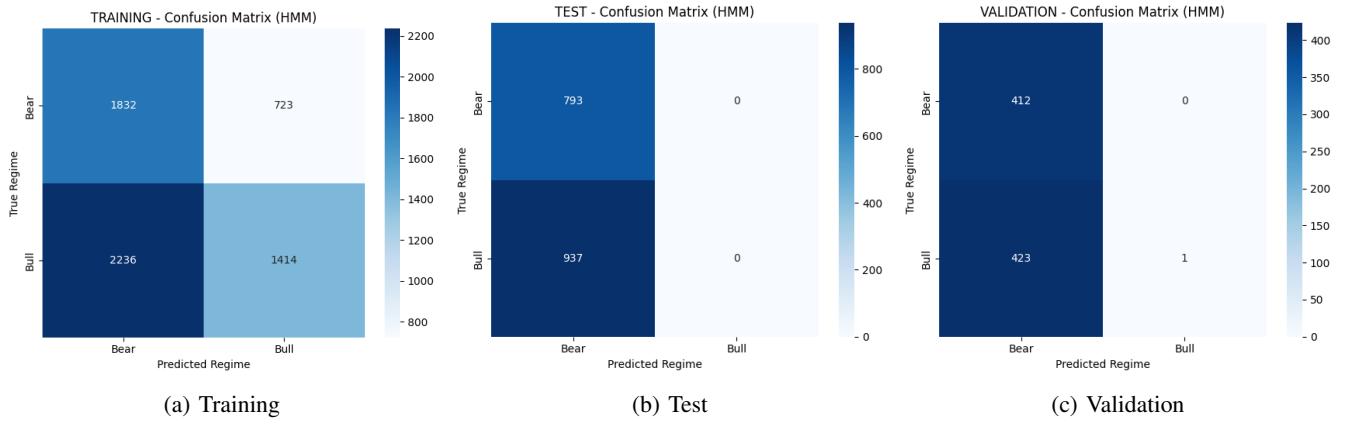


Fig. 5: HMM confusion matrices for market regime detection across training, test, and validation sets.

its 54.6 percent accuracy and 0.380 F1-score indicate that NVT alone is not reliable for classifying overvalued and undervalued cryptocurrency assets.

V. COMPARATIVE ANALYSIS

Table IX compares the proposed framework with related studies. Unlike prior work that focuses mainly on Bitcoin or single prediction tasks, this study evaluates multiple research questions across Bitcoin and Ethereum. The framework achieves strong results in volatility prediction, speculative behavior classification, and adoption phase detection while also identifying the limitations of market cycle and NVT-based valuation models.

TABLE IX: Comparison with Related Works

Work	Problem	Model	Data	Best Result
King et al. [4]	Trading indicators	On-chain metrics	BTC	Varied
Berger et al. [5]	Return forecasting	ML, econometric	BTC	ML better
Huang et al. [7]	Volatility	ML models	BTC	Improved forecast
Guo et al. [6]	Cross-crypto returns	Adaptive LASSO	Multi	OOS profit
This work	Adoption, volatility, behavior, valuation	RF, LR, LGBM, XGB, HMM	BTC+ETH	96.81%

The strongest performance is obtained for volatility prediction and adoption classification because these tasks align closely with on-chain activity. In contrast, return direction, HMM market cycles, and NVT valuation are weaker because they depend on broader market forces, external events, and investor behavior that are not fully captured by on-chain variables. Overall, the proposed framework contributes a broader evaluation than prior single-task studies by showing where on-chain analytics are reliable and where they remain insufficient.

VI. CONCLUSION AND FUTURE WORK

This paper presented an analytical framework for measuring cryptocurrency adoption, network activity, and market behavior using Bitcoin and Ethereum on-chain and market data. The

results show that Daily Active Addresses, transaction fees, hash rate, and transaction behavior are useful for adoption and volatility analysis. Linear Regression performed poorly for return prediction, achieving a negative R^2 score of -3.2908 and a high RMSE of 1.055×10^6 . Logistic Regression achieved 96.81 percent accuracy for volatility classification, LightGBM with SMOTE achieved 93.48 percent accuracy for distinguishing genuine and speculative behavior, and Random Forest achieved 95.20 percent accuracy for adoption phase classification. However, on-chain metrics were weak predictors of short-term return direction, while HMM and NVT-based valuation showed limited reliability.

Future work should include more cryptocurrencies, longer time horizons, proof-of-stake indicators, and external macroeconomic or sentiment features. Transformer-based temporal models and causal inference methods may further improve market cycle detection and clarify how on-chain adoption affects financial outcomes. A real-time dashboard could also be developed to support researchers, investors, and policymakers in monitoring cryptocurrency adoption and risk.

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